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18 April 1986
SUPERSEDING
QQ-P-416D
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FEDERAL SPECIFICATION

PLATING, CADMIUM (ELECTRODEPOSITED)

This specification is approved by the Commissioner, Federal Supply Service, General Services Adminstration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for electrodeposited cadmium plating. Any process of cadmium deposition should not be used when an alternate process meets the performance requirements of this specification and is considered satisfactory for use on the item under consideration (See 6.2.4).

1.2 Classification.

1.2.1 Classes. Cadmium plating shall be of the following classes, as specified (see 6.2):

Class 1 - 0.00050 inch thick.
Class 2 - .00030 inch thick.
Class 3 - .00020 inch thick.

1.2.2 Types. Cadmium Plating shall be of the following types, as specified (see 6.2):

Type I - As plated.
Type II - With supplementary chromate treatment (see 3.2.9)
Type III - With supplementary phosphate treatment (see 3.2.10)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

AREA MFFP

AMSC N/A

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Federal Specifications:

QQ-S-624	-	Steel Bar, Alloy, Hot Rolled and Cold Finished (General Purpose).
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

Federal Standard

Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-S-5002	-	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1312	-	Fasteners, Test Methods.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

B117 - Salt Spray (Fog) Testing
B487 - Measuring Metal and Oxide Coating Thickness by Microscopic Examination
B499 - Measuring Coating Thickness by Magnetic Method-Nonmagnetic Coatings on Magnetic Basis Metal
B504 - Measurement of the Thickness of Metallic Coatings by the Coulometric Method
B529 - Measurement of Coating Thickness by the Eddy Current Test Method; Nonconductive Coatings on Nonmagnetic Basis Metals
B567 - Test for Measuring Coating Thickness by the Beta-Back-scatter Principle"
B568 - Test for Measuring Coating Thickness by X-Ray Spectrometry
E8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Materials. The materials used shall be such as to produce platings which meet the requirements of this specification.

3.2 General requirements.

3.2.1 High tensile strength steel parts. Unless otherwise specified, steel parts having an ultimate tensile strength greater than 240,000 pounds per square inch (psi) shall not be plated without specific approval of the procuring activity (see 6.2).

3.2.2 Stress relief treatment. Unless otherwise specified, all steel parts having an ultimate tensile strength of 150,000 pounds per square inch (psi) and above, which are machined, ground, cold formed or cold straightened, after heat treatment, shall be baked at a minimum of 375 ± 25°F (191 ± 14°C) for three hours or more prior to cleaning and plating for the relief of damaging residual tensile stresses (see 6.2).

3.2.3 Cleaning. Unless otherwise specified, all steel parts shall be cleaned in accordance with MIL-S-5002 (see 6.2).

3.2.4 Plating application. Unless otherwise specified, the plating shall be applied after all basis metal heat treatments and mechanical operations, such as machining, brazing, welding, forming and perforating of the article have been completed (see 6.2).

3.2.5 Underplating. Unless otherwise specified, cadmium shall be deposited directly on the basis metal without a preliminary plating of other metal, such as nickel, or copper, except in the case of parts made of corrosion-resistant alloys on which a preliminary plating of nickel is permissible or of parts made of aluminum on which a preliminary treatment such as the zincate process is permissible (see 6.2).

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3.2.6 Coverage. Unless otherwise specified, the plating shall cover all surfaces as stated in 3.3.1, 3.3.1.1 and 3.3.1.2 including roots of threads, corners and recesses (see 6.2).

3.2.7 Luster. Unless otherwise specified, either a bright or dull luster shall be acceptable (see 6.2). However, parts for aircraft and aerospace applications, that are heat treated to Rockwell C40 and above, shall have a dull luster or finish.

3.2.8 Embrittlement relief. Unless otherwise specified or stated in the end product specifications, all steel parts having a hardness of Rockwell C40 and higher shall be baked at a minimum of 375 ± 25°F (191 ± 14°C) for three hours or more, within four hours after plating, to provide hydrogen embrittlement relief (see 6.6). The baked parts, when tested in accordance with 4.5.4, shall not crack or fail by fracture (see 4.4.3.4). Plated springs and other parts subject to flexure shall not be flexed prior to hydrogen embrittlement relief treatment. In the case of types II and III treated parts, the baking treatment shall be done prior to the application of the supplementary coatings. Cadmium plated surfaces passivated as a result of the baking operation shall be reactivated prior to receiving the type II supplementary chromate treatment (see 6.9).

3.2.9 Chromate treatment (Type II). Unless otherwise specified in the contract or order (see 6.2), the chromate treatment required for conversion to Type II shall be a treatment in or with an aqueous solution of salts, acids, or both, to produce a continuous smooth, distinct protective film, distinctly colored iridescent bronze to brown including olive drab and yellow. The articles so treated shall be thoroughly rinsed and dried in accordance with the requirements of the process used. Type II plating shall be similar in appearance to platings on separate specimens which are capable of passing the salt-spray test (see 3.3.4, 4.4.3.3, and 4.5.3). Usual chromic and nitric acid bright dips for cadmium are not chromate treatments.

3.2.10 Phosphate treatment (Type III). Unless otherwise specified in contract or order the phosphate treatment required for conversion to type III shall produce a tightly adherent film conforming to type I of TT-C-490 (see 6.2).

3.3 Detail requirements.

3.3.1 Thickness of plating. Unless otherwise specified in the contract or order, (see 6.2), the thickness of cadmium for other than fastener hardware (see 3.3.1.1) shall be as specified in Table I on all visible surfaces which can be touched by a ball 0.75 inch in diameter. Where Class 1 is specified, all other visible surfaces shall be Class 2 minimum thickness. If the maximum thickness for Class 1 is not specified in the contract, order or applicable drawing, the thickness shall not exceed 0.0008 inch (0.8 mil). Where Class 2 is specified, all other visible surfaces shall be Class 3 minimum thickness. Where Class 3 is specified, all other visible surfaces shall be not less than 0.00015 inch minimum thickness.

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3.3.1.1 For fastener hardware. Unless otherwise specified in the end product specifications, fastener hardware shall have Class 3 thickness plating. (see 4.5.1 and 6.5). There shall be no bare areas.

3.3.1.2 For other than fastener hardware. For other than fastener hardware, the cadmium plating shall be Class 1 thickness unless otherwise specified in the contract or order or controlled by the following exceptions (see 6.2):

(a) Articles with portions externally threaded shall have a minimum of class 2 thickness on the threaded portions
(b) Holes and other openings and articles with internal threads from which the external environment is completely excluded shall not be subjected to thickness requirements but shall show evidence of coating. There shall be no bare areas.

TABLE I. Thickness

Class	Thickness	
	Inch minimum	Equivalent thickness micrometers (approx) 1/
1	0.0005	13
2	0.0003	8
3	0.0002	5

1/ 0.001 inch = 1 mil = 25.4 micrometers (microns).

3.3.2 Types. Unless otherwise specified in the contract or order (see 6.2), the cadmium plating shall be Type II. For use on surfaces to be painted, the cadmium plating shall be either Type II or Type III (see 6.1.2 and 6.1.3).

3.3.3 Adhesion. The adhesion of the plating shall be such that when examined at a magnification of approximately 4 diameters, the plating shall not show separation from the basis metal or from any underplating at the interface, nor shall any underplate show separation from the basis metal at the interface when subjected to the tests described in 4.5.2. The interface between the plating and either the basis metal or the underplate is the surface before plating. The interface between the underplate and the basis metal is the surface before underplating. The formation of cracks in the plating caused by rupture of the basis metal, the underplate or combination of both which do not result in flaking, peeling or blistering of the plating shall not be considered as nonconformance to this requirement.

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3.3.4 Corrosion resistance. Cadmium plating with the Type II treatment shall show neither white corrosion products of cadmium nor basis metal corrosion products at the end of 96 hours when tested by continuous exposure to the salt spray in accordance with 4.5.3. The appearance of corrosion products, visible to the unaided eye at normal reading distance shall be cause for rejection, except that white corrosion products at the edges of specimens shall not constitute failure.

3.4 Workmanship.

3.4.1 Basis metal. The basis metal shall be free from visible defects that will be detrimental to the appearance or protective value of the plating. The basis metal shall be subject to such cleaning and plating procedures as necessary to yield platings herein specified.

3.4.2 Plating. The cadmium plating shall be smooth, fine grained, adherent, uniform in appearance, free from blisters, pits, nodules, burning and other defects. The plating shall show no indication of contamination or improper operation of equipment used to produce the cadmium deposit, such as excessively powdered or darkened platings. Superficial staining which has been demonstrated as resulting from rinsing or slight discoloration resulting from any drying or baking operation as specified (see 3.2.8) shall not be cause for rejection. All details of workmanship shall conform to the best practice for quality plating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Production control inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Production control inspection.

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4.3.1 Control records. When specified in the contract or order (see 6.2), the supplier shall maintain a record of each processing bath, showing all additional chemicals or treatment solutions to the unit, the results of all analyses performed and the quantity of parts plated during operation. Upon request of the procuring activity, such records shall be made available. These records shall be maintained for not less than one year after completion of the contract or purchase order.

4.3.2 Production control. The equipment, procedures and operations employed by a supplier shall be capable of producing high quality electrodeposited cadmium plating on materials as specified in this document. When specified by the procuring activity (see 6.2), the supplier, prior to production, shall demonstrate the capability of the process used to show freedom from hydrogen embrittlement damage as indicated by satisfactory behavior of specimens prepared (see 6.2.2) and tested in accordance with 4.3.2.1 to comply to the requirements of MIL-S-5002 for preproduction process qualification.

4.3.2.1 Preproduction control. For preproduction control four round notched steel specimens shall be prepared in accordance with 4.4.4.3 from four individual heats for a total of 16 specimens, using the specified steel alloy for which preproduction examination of the process is to be demonstrated. Specimens shall be heat treated to the maximum tensile strength representing production usage. The specimens shall be given the same pre-treatments, electroplating and post-plating treatments proposed for production. The specimens shall be subject to test detailed in 4.5.4. The process shall be considered satisfactory if all specimens show no indication of cracks or failure. The tests results and production control information shall be submitted to the procuring activity for approval. Until approval has been received, parts shall not be plated.

4.3.3 Frequency of tests. To assure continuous control of the process as required by MIL-S-5002 and to prevent detrimental hydrogen embrittlement during production, the satisfactory behavior of specimens, prepared and tested in accordance with Table II, shall be made once each month or more frequently if required by the procuring activity. The results of tests made to determine conformance of electrodeposited platings to all requirements of this specification for definite contracts or purchase order are acceptable as evidence of the properties being obtained with the equipment and procedures employed.

4.3.4 Production control specimens. Test specimens for production control shall be prepared in accordance with 4.4.4, 4.4.4.1 and 4.4.4.2 as applicable for the thickness, adhesion and corrosion resistance tests detailed in Table II. Specimens for the production control embrittlement relief test shall be four round notched steel specimens of alloy steel 4340 conforming to QQ-S-624, heat treated to the maximum tensile strength, from one or more heats, and prepared in accordance with 4.4.4.3.

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Table II. Production Control Tests and Specimens

Test	For coating types	Requirement paragraphs	Specimen preparation paragraphs	Test reference paragraphs
Thickness	I, II, III	3.3.1, 3.3.1.1 and 3.3.1.2	4.4.4 and 4.4.4.1 1/	4.5.1
Adhesion	I, II, III	3.3.3	4.4.4 and 4.4.4.1 1/	4.5.2
Corrosion resistance	II	3.3.4	4.4.4 and 4.4.4.2 1/	4.5.3
Hydrogen embrittlement	I, II, III	3.2.8	4.3.4, 4.4.4, and 4.4.4.3	4.5.4

1/ Standard alloy steels shall be used for production control specimens. The selection shall be at the option of the supplier; however alloy steels such as AISI or SAE numbers 4130, 4135, 4140, 4145, 4340, 8645 and 8740 conforming to QQ-S-624 shall be used.

4.4 Quality conformance inspection

4.4.1 Lot. A lot shall consist of plated articles of the same basis metal composition, class and type plated and treated under the same conditions and approximately the same size and shape submitted for inspection at one time.

4.4.2 Sampling for visual examination and non destructive tests. Sampling for visual examination and non destructive tests shall be conducted at the option of the supplier in accordance with MIL-STD-105 or using Table III. A sample of coated parts or articles, except for those barrel plated, shall be drawn by taking a random from each lot the number of articles in accordance with MIL-STD-105, Level II, Acceptable Quality Level (AQL) 1.5 percent defective, or as indicated in Table III. Barrel plated parts or articles shall be sampled in accordance with Level S-3 of MIL-STD-105, Acceptable Quality Level (AQL) 4.0 percent defective. The lot shall be accepted or rejected according to the procedures in 4.4.2.1 for visual examination and 4.4.2.2 for plating thickness (non destructive tests).

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TABLE III. Sampling for visual examination and nondestructive tests

Numbers of items in lot inspections	Number of items in samples (randomly selected)	Acceptance number (maximum number of sample items nonconforming to any test)
15 or less	7 1/	0
16 to 40	10	0
41 to 110	15	0
111 to 300	25	1
301 to 500	35	1
501 and over	50	2

1/ If the number of items in the inspection lot is less than 7, the number of items in the sample shall equal the number of items in the inspection lot.

4.4.2.1 Visual examination. Samples selected in accordance with 4.4.2 shall be examined for compliance with requirements of 3.4.2 after plating. If the number of non conforming articles exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.2.2 Thickness of plating (nondestructive tests). Samples selected in accordance with 4.4.2 shall be inspected and the plating thickness measured by the applicable tests detailed in 4.5.1, at several locations on each article as defined in 3.3.1, 3.3.1.1 or 3.3.1.2, as applicable, for compliance with the requirements. Measurements on fastener hardware (see 3.3.1.1) shall be made at locations defined in MIL-STD-1312, Test 12. The part or article shall be considered nonconforming if one or more measurements fail to meet the specified minimum thickness. If the number of defective items in any sample exceeds the acceptance number for the specified sample, the lot represented by the sample shall be rejected. Separate specimens (see 4.4.4.1) shall not be used for thickness measurements unless a need has been demonstrated.

4.4.3 Sampling for destructive tests. A random sample of four plated parts or articles shall be taken from each lot for each destructive test or separately plated specimens shall be prepared in accordance with 4.4.4, 4.4.4.1, 4.4.4.2 and 4.4.4.3 to represent each lot. If the number of articles in the lot is four or less, the number of articles in the sample shall be specified by the procuring activity (see 6.2).

4.4.3.1 Thickness of plating (destructive tests). If sampling and testing for thickness of plating by nondestructive testing is not the option of the supplier, samples selected in accordance with 4.4.3 shall be measured for

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plating thickness by the applicable test detailed in 4.5.1, at several locations on each article as defined in 3.3.1, 3.3.1.1 or 3.3.1.2 for compliance with the requirements. Measurements on fastener hardware (see 3.3.1.1) shall be made at locations defined in MIL-STD-1312, Test 12. If the plating thickness at any place on any article or specimen is less than the specified minimum thickness, the lot shall be rejected. Separate specimens (see 4.4.4.1) shall not be used for thickness measurements unless a need has been demonstrated.

4.4.3.2 Adhesion (destructive tests). The articles or specimens used for the destructive thickness test (see 4.4.3.1), if of suitable size and form, may be used as the test pieces for the adhesion test to determine compliance with the requirements of 3.3.3. Failure of one or more of the test pieces shall constitute failure of the lot.

4.4.3.3 Corrosion resistance (destructive tests). When specified in the contract or order, compliance with the requirements for corrosion resistance of Type II treated articles shall be determined (see 6.2). A set of four separate test specimens, prepared in accordance with 4.4.4 and 4.4.4.2 in lieu of the treated coated articles shall be used to determine compliance with the requirements for corrosion resistance (see 3.3.4). Failure of one or more of the test specimens shall reject the lot.

4.4.3.4 Hydrogen embrittlement relief (destructive tests). When specified in the contract or order, conformance to the requirements of 3.2.8 for hydrogen embrittlement relief of treated parts shall be determined for those parts which will be subject to a sustained tensile load in use (see 6.2). A random sample of four plated articles shall be taken from each lot of four specimens, prepared in accordance with 4.4.4 and 4.4.4.3, shall be used to represent the lot. When tested as specified in 4.5.4, cracks or failure by fracture shall be cause for rejection. Failure of one or more of the test pieces shall reject the lot.

4.4.4 Quality conformance specimen preparation. When the plated articles are of such form, shape, size and value as to prohibit use thereof, or are not readily adaptable to a test specified herein, or when destructive tests of small lot sizes are required the test shall be made by the use of separate specimens plated concurrently with the articles represented. The separate specimens shall be of a basis metal equivalent to that of the articles represented. "Equivalent" basis metal includes chemical composition, grade, condition and finish of surface prior to plating. For example, a cold-rolled steel surface should not be used to represent a hot-rolled steel surface. Due to the impracticality of forging or casting separate test specimens, hot-rolled steel specimens may be used to represent forged and cast-steel articles. The separate specimens may be also cut from scrap castings when ferrous alloy castings are being plated. These separate specimens shall be introduced into a lot at regular intervals prior to the cleaning operations, preliminary to plating, and shall not be separated therefrom until after completion of plating. Conditions affecting the plating of specimens

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including the spacing and plating media, bath agitation, temperature, etc. in respect to other objects being plated shall correspond as nearly as possible to those affecting the significant surfaces of the articles represented. Separate specimens shall not be used for thickness measurements, however, unless the necessity for their use has been demonstrated.

4.4.4.1 Specimens for thickness and adhesion tests. If separate specimens for thickness and adhesion tests are required, they shall be strips approximately 1 inch wide, 4 inches long and 0.04 inch thick.

4.4.4.2 Specimens for corrosion resistance tests. If separate specimens for corrosion resistance test are required, they shall be panels not less than 6 inches in length, 4 inches in width and approximately 0.04 inch thick.

4.4.4.3 Specimens for embrittlement relief. Separate specimens for embrittlement relief test shall be round notched specimens with the axis of the specimen (load direction) perpendicular to the short transverse grain flow direction. The configuration shall be in accordance with figure 8 of ASTM E8 for rounded specimens. Specimens shall have a 60 degree V-notch located approximately at the center of the gage length. The cross section area at the root of the vee shall be approximately equal to half the area of the full cross section area of the specimen's reduced section. The vee shall have a 0.010 $\pm$ 0.0005 radius of curvature at the base of the notch (see 6.2.2).

4.5 Tests.

4.5.1 Thickness. For nondestructive measuring of plating thickness, procedures in accordance with Fed. Test Method Std. No. 151, Method 520 (electronic test), ASTM B499 (magnetic test), ASTM B529 (eddy current), ASTM B567 (test by beta radiation back scatter principle) or ASTM B568 (X-ray spectrometry) may be used. For destructive measuring of plating thickness, procedures in accordance with ASTM B487 (microscopic) or ASTM B504 (coulometric) may be used. In addition to the above, the other procedures embodied in MIL-STD-1312, Test 12 may be used for thickness measurement of plating fastener hardware. Thickness measurements of cadmium platings, Types II and III, shall be made after application of the supplementary treatments. When the coulometric test is used, the supplementary treatment shall be removed prior to testing. The chromate film may be removed from Type II coating by using a very mild abrasive (a paste of levigated alumina rubbed on with the finger). The phosphate coating may be removed from the type III coating by immersing the specimen in a 10 percent solution of NaOH and scrubbing with a rubber policeman (usually takes from 10 to 15 minutes).

4.5.2 Adhesion. Adhesion may be determined by scraping the surface or shearing with a sharp edge, knife, or razor through the plating down to the basis metal and examining at four diameters magnification for evidence of non-adhesion. Alternately, the article or specimen may be clamped in a vise and the projecting portion bent back and forth until rupture occurs. If the edge of the ruptured plating can be peeled back or if separation between the plating and basis metal can be seen at the point of rupture when examined at four diameters magnification, adhesion is not satisfactory.

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4.5.3 Corrosion resistance. Corrosion resistance shall be conducted in accordance with ASTM B117 (salt spray test) for 96 hours or in accordance with MIL-STD-1312, Test 1 for fastener hardware. To secure uniformity of results, type II supplementary coatings shall be aged at room temperature for 24 hours before subjection to the salt spray.

4.5.4 Embrittlement relief. Compliance with 3.2.8 shall be determined with samples of plated parts taken as specified in 4.4.3.4. Parts such as spring pins, lock rings, etc., which are installed in holes or rods shall be similarly assembled using the applicable parts specifications or drawing tolerances which impose the maximum sustained tensile load on the plated part. The selected samples shall be subjected to a sustained tensile load equal to 115 percent of the maximum design yield load for which the part was designed. Fastener hardware, where the maximum design yield load is not known or given, shall be tested in accordance with MIL-STD-1312, Test 5. Parts which require special fixtures, extreme loads to comply with the above requirements, or where the maximum design yield load is not known, may be represented by separate specimens prepared in accordance with 4.4.4.3. The notched specimens shall be subject to a sustained tensile load equal to 75 percent of the ultimate notch tensile strength of the material. The articles, parts or specimens shall be held under load for at least 200 hours and then examined for cracks or fracture.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Preservation, packaging and packing methods for electrodeposited cadmium plated parts or articles employed by a supplier shall be such as to preclude damaging during shipment and handling.

6. NOTES

6.1 Intended use.

6.1.1 Use. The electrodeposited cadmium platings covered by this specification are intended for use as corrosion protective platings on ferrous and other basis metal parts. On ferrous parts heat-treated or having an ultimate strength between 180,000 to 240,000 psi (approximately Rockwell C40 to C49), such processing should be used for cleaning and cadmium deposition by electroplating that little or no hydrogen embrittlement will result. On steel heat-treated or having an ultimate strength greater than 240,000 psi, cadmium deposition by electrodeposition is not recomended.

6.1.2 Type II treatment. The prime purpose of chromate finishes (type II) on electrodeposited cadmium platings is to retard or prevent the formation of white corrosion products on surfaces exposed to stagnant water, high humidity atmospheres, salt water, marine atmospheres or cyclic condensation and drying. Some types of chromate coatings have proved satisfactory as a base for paints.

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6.1.3 Type III treatment. The prime purpose of phosphate finishes (type III) on electrodeposited cadmium platings is to form a paint base.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification

(b) Class and Type required (see 1.2.1, 1.2.2, and 3.3.2)

(c) When plating is to be applied, if other than specified (see 3.2.1, 3.2.4, and 3.2.6)

(d) Stress relief treatment, if other than specified (see 3.2.2)

(e) Cleaning of steel, if other than specified (see 3.2.3)

(f) Underplating required (see 3.2.5)

(g) Luster required, if other than specified (see 3.2.7)

(h) Hydrogen embrittlement relief treatment, if other than specified (see 3.2.8)

(i) Type of chromate treatment required for conversion to type II, if other than specified (see 3.2.9)

(j) Type of phosphate treatment required for conversion to type III, if other than specified (see 3.2.10)

(k) Thickness of coating, if other than specified (see 3.3.1, 3.3.1.1 and 3.3.1.2)

(l) Control record requirement (see 4.3.1)

(m) Preproduction control examination (see 4.3.2)

(n) Number of samples for destructive testing (see 4.4.3)

(o) Whether corrosion resistance test is required (see 4.4.3.3)

(p) Whether hydrogen embrittlement relief test is required (see 4.4.3.4)

6.2.1 The manufacturer of the basis metal parts should provide the plating facility with the following data:

(a) Hardness of steel parts (see 3.2.1)

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(b) Heat treatment for stress relief, whether it has been performed or is required (see 3.2.2)

(c) Tensile loads required for embrittlement relief test, if applicable (see 4.5.4).

6.2.2 The manufacturer of the basis metal parts should provide the plating facility with notched tensile specimens (see 4.4.4.3) to be plated for conformance with 3.2.8 required for production control (see 4.3.2.1, and 4.3.4) and lot acceptance (see 4.4.3 and 4.4.3.4).

6.2.3 When QQ-P-416 is specified for articles to be surface finished, coatings of cadmium in accordance with MIL-C-81562, Coatings, Cadmium and Zinc (Mechanically Deposited), may be furnished in lieu of the electrodeposited coating at the option of the supplier, within the limitations of this specification and upon acceptance and approval of the procuring activity.

6.2.4 Information relative to an alternate process should be furnished to the cognizant Government activity for approval prior to use on the item under consideration.

6.3 Limitations. Cadmium platings should not be used on parts for space applications or on parts which in service reach a temperature of 450°F (233°C) or higher or come in contact with other parts which reach those temperatures. MIL-S-5002 contains additional warning where the cadmium plating shall not be used for weapons systems applications.

6.4 Stress relief. There is a hazard that hardened and tempered cold-worked or cold-straightened steel parts may crack during cleaning and plating. Such parts should have a suitable heat treatment for stress relief prior to cleaning and plating (see 3.2.2).

6.5 Threaded parts. As heavier platings are required for satisfactory corrosion resistance than class 3 for military use, allowance should be made in the manufacture of most threaded articles, such as nuts, bolts and similar fasteners with complementary threads for dimensional tolerances to obtain necessary coating build-up. Certain recessed areas, such as root diameters of threads, have a tendency to exhibit lack of build-up with electrodeposited platings and vacuum deposited cadmium in contrast with mechanically deposited cadmium coatings.

6.6 Baking time. For high strength materials (Rockwell C40 and above), it may be beneficial to extend the baking time to 23 hours to insure complete hydrogen embrittlement relief.

6.7 Type II temperature limitations. Chromate treatments (Type II) should not be used on plated parts that will not be painted and which will be continuously exposed to temperatures in excess of 150°F (66°C) or intermittently exposed for short periods to temperatures of approximately 300°F (131°C) or more. However, these treatments may be used to prevent finger marking and corrosion which may occur at room temperature during assembly and storage.

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6.8 Type II handling precaution. Chromate treatments (type II), which involve only dipping in chemical solutions, normally require a sufficient period of drying, approximately 24 hours at 70 to 90°F (21 to 32°C) to render the parts suitable for handling without damage to the coating while in gelatinous forms; and it is important with such coatings that the workmanship be such that the coating is not excessively damaged while wet.

6.9 Reactivation. Surfaces of cadmium plating intended for conversion to type II and which have become passive as a result of the baking operation specified in 3.3.6 may be reactivated by brief immersion in dilute acid. If, for example, the chromating solution is acidified with sulfuric acid, then the reactivating solution should be 1 part sulfuric acid (sp. gr. 1.83) by volume added to 99 parts water, or for further example, if the chromating solution is acidified with hydrochloric acid, then the activating solution should be 1 part hydrochloric acid (sp. gr. 1.16) by volume added to 99 parts water. The duration of immersion should be as brief as is consistent with the nature of the work. For example, a perforated container of barrel-plated parts would be expected to be reactivated in approximately 15 seconds and separately racked items in approximately 5 seconds. The surfaces should be activated as soon as possible following baking operation and should be handled carefully to avoid contamination by dirt or grease.

6.10 Toxicity. Cadmium, because of its toxicity, shall not be employed as a plating for any object intended for use as a food container or cooking utensil or for any object likely to come in contact with food. Cadmium plated sheets and any other structural shapes which may be subjected to heat from welding, brazing or soldering operations should be so labeled because of the danger from poisonous vapors during operations.

6.11 Packaging limitations. Unprotected cadmium plated articles should not be packed in non-ventilated containers, either together or in contact with electrical equipment, because of the danger of deleterious effect on cadmium plating from unstable organic electrical insulation. In addition to organic electrical insulation, phenolic resinous substances and other containing unsaturated carbon-to-carbon linkages, such as oil paints and impregnated paper, etc., cause an abnormal attack on cadmium by setting free in the presence of moisture formic acid, butyric acid, etc. Corrosion of cadmium platings and steel basis metal has been noted when cadmium plated articles have been packaged in direct contact with container materials such as wood or cardboard. Corrosion has been especially severe if the container materials have become wet or have been stored under conditions of high humidity.

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MILITARY INTEREST:

Custodians

Army - MR
Navy - AS
Air Force - 20

Review Activities

Army - AL, AR, ER
Navy - EC, OS
Air Force - 70, 99
DLA - IS(E)

User Activities

Army - AT, ME, MI
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
COM - NBS

PREPARING ACTIVITY:

NAVY - AS

DOD Project MFFP-0373

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
QQ-P-416E	PLATING, CADMIUM (ELECTRODEPOSITED)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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